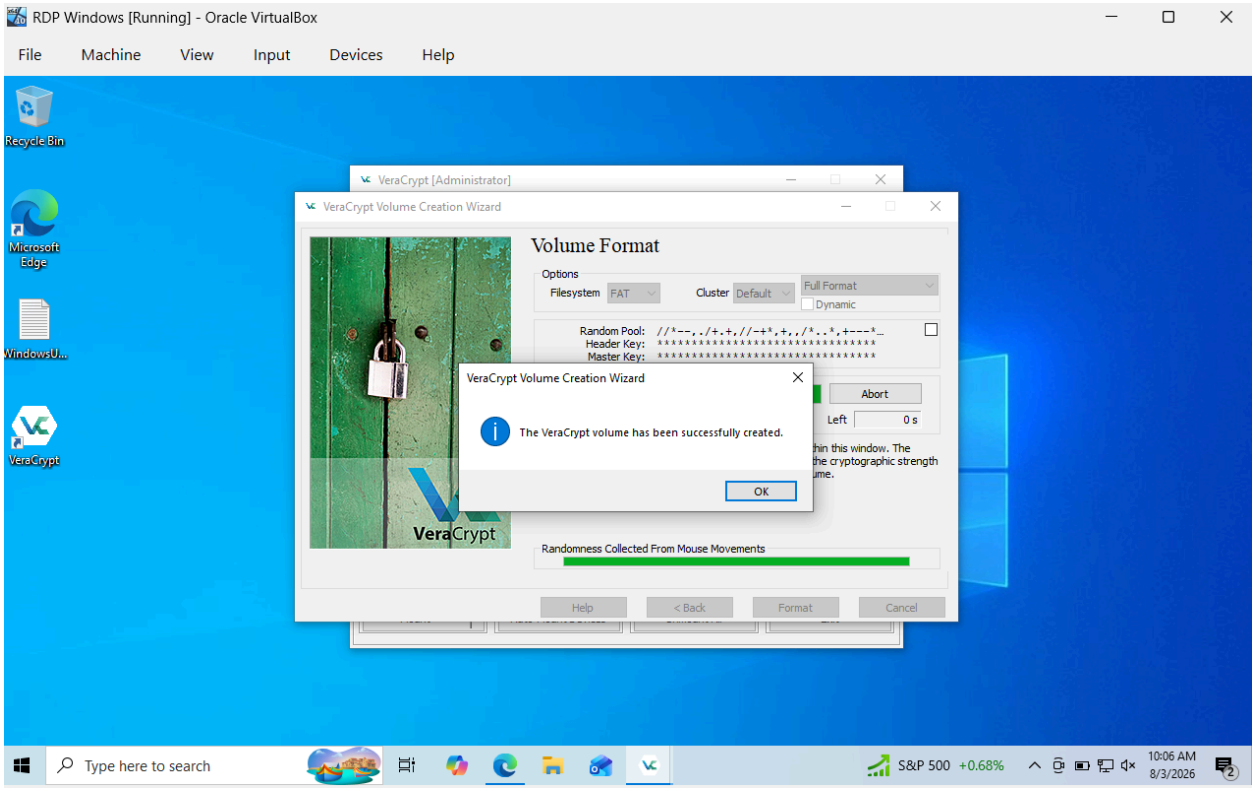
Frank Henry

Disk Encryption

08/03/2026

Disk Encryption Using VeraCrypt



VeraCrypt is a third party encryption program that allows us to encrypt a folder on our computer's hard drive. Once the folder is encrypted, anything that goes into it gets automatically encrypted, and cannot be decrypted without the password, or unless the file is moved from that folder.

Reflection Questions

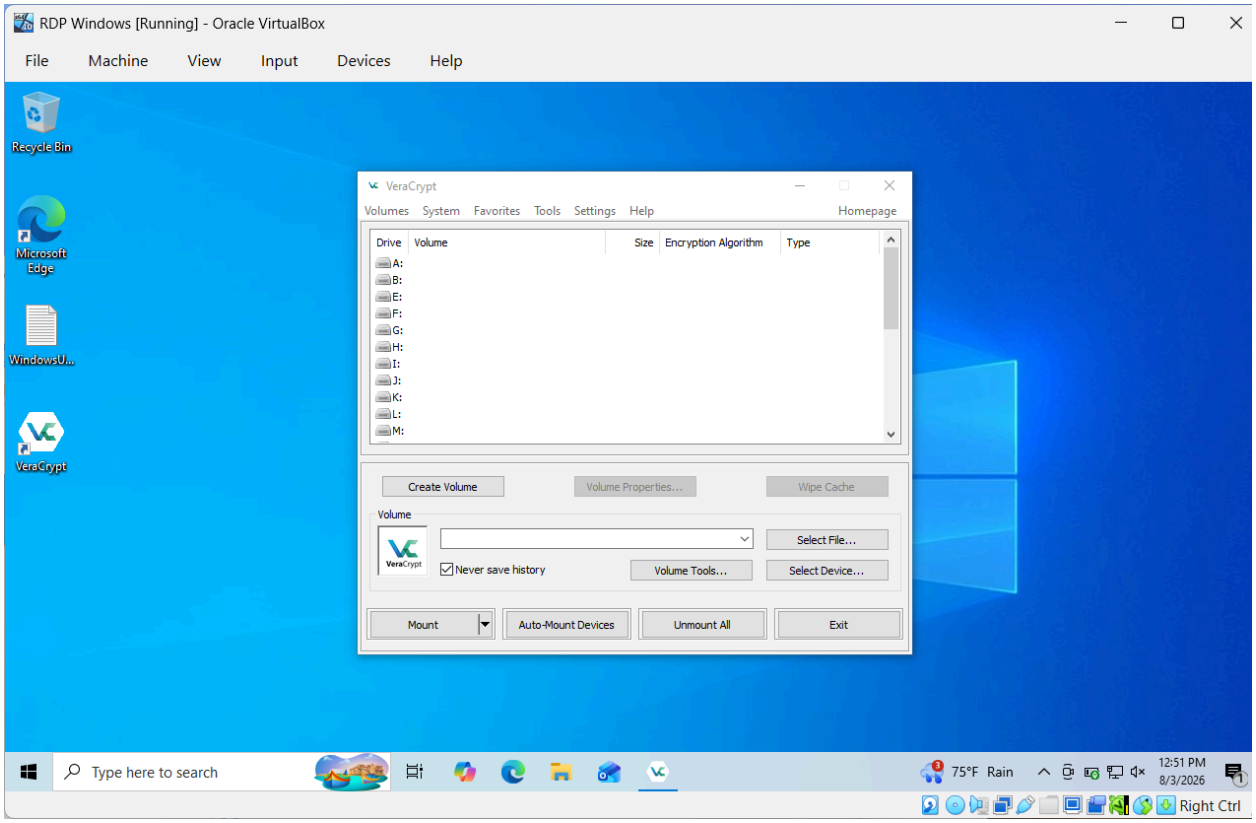
1. What are the main differences between **VeraCrypt** and **BitLocker** in terms of functionality and use cases?
    - Due to not having BitLocker installed on either my Virtual Machine, nor main PC, I was unable to use and test BitLocker as a program. But after doing some research and using VeraCrypt, I was still able to figure out the main differences of both programs. VeraCrypt protects anything you put

into the created folder that is placed onto your PC and password protected. BitLocker is useful for your entire PC. It can lock up your PC with an access code and the entire PC will be encrypted rather than just a single folder. BitLocker is also used solely for Windows, but VeraCrypt is a cross-platform program that can be used on any OS from Windows to Mac, to Linux, etc.

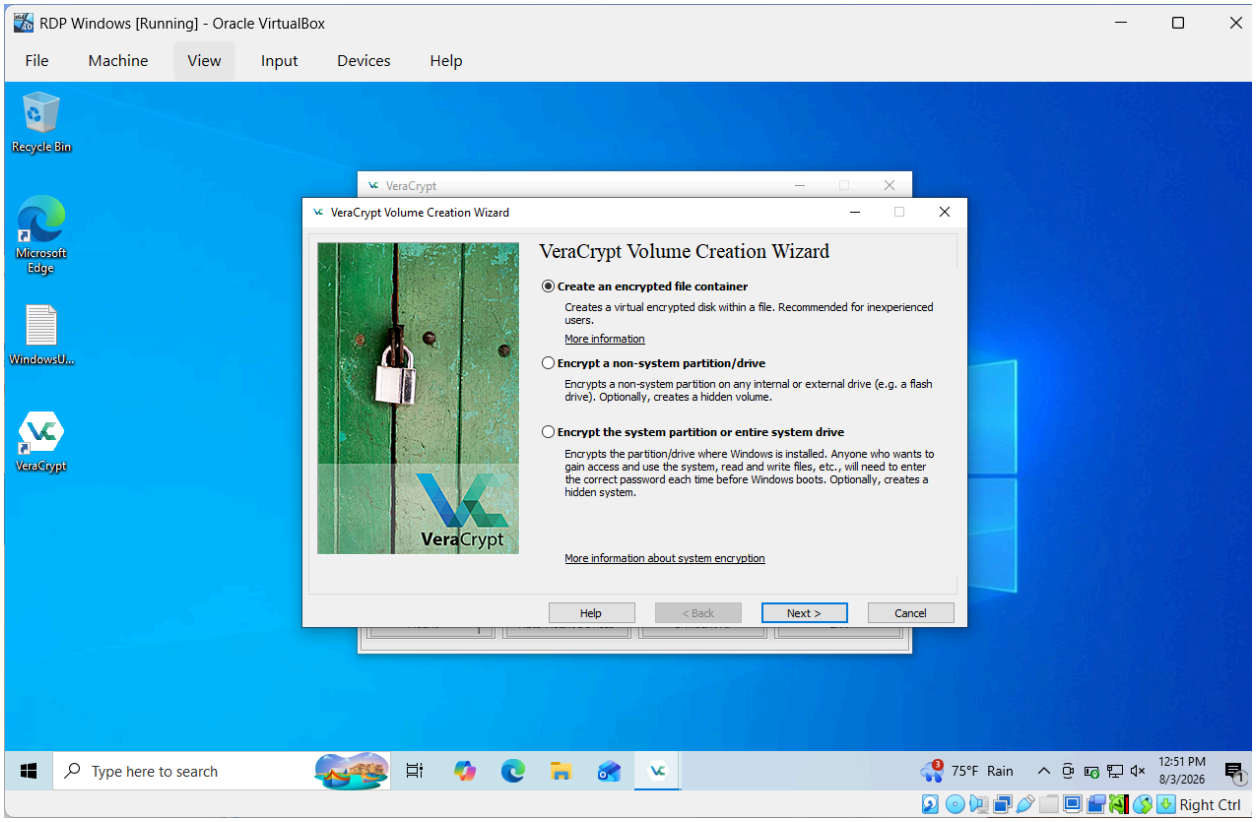
2. What are the security benefits of encrypting a full disk compared to only encrypting individual files?
    - The security benefits are pretty self explanatory in my opinion. Encrypting a full disk is highly recommended if the work you do is mostly classified, private work, like government work for example. If you use VeraCrypt for private work, there's always a possibility that as humans, we may forget to place a secret file into a VeraCrypt folder. But if BitLocker is turned on a PC, the entire disk is encrypted in the case of any potential threats trying to get into your PC.
3. What might happen if you lose your encryption key or password?
    - Both programs are the same for this matter with one main difference. If you forget your password AND your recovery key, you lose all the data, forever. There's one main difference between the 2 though. BitLocker usually saves your key to your Microsoft account automatically. Or you can manually use a usb drive, text file, or printed copy. VeraCrypt does not automatically back up the key anywhere, so unless manually saved elsewhere, the data will be encrypted forever.
4. Why should encryption always be tested inside a **virtual environment** first?
    - Just in case you encrypt on your main PC and lose the key, your PC data is gone forever. If you test on a VM that doesn't have any sensitive information, you can just completely erase that machine and recreate an entire new one. No harm no foul in a VM.
5. Which tool would you recommend for enterprise use and why?
    - When it comes to enterprise use, I would recommend using BitLocker. At least in this instance, it can be controlled and monitored by the IT

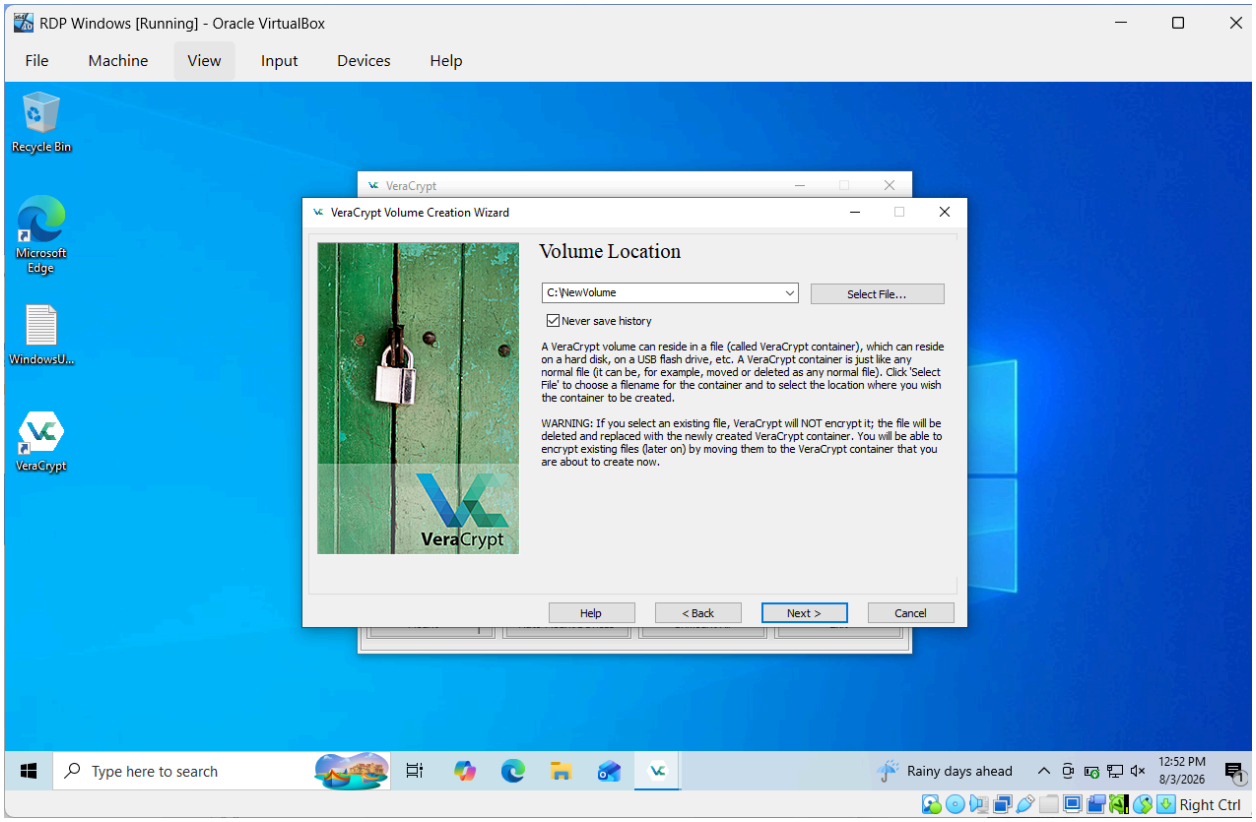
department. It would be controlled centrally so they can enforce encryption policies and key storage recoveries to prevent any hardcore damage to systems. There are also common enterprise compliance frameworks like FIPS 140-2 that many industries require.

- BitLocker is included with most of the Operation Systems that companies use, like Microsoft Enterprise. So there's no additional cost to the company for encryption protection since they're already paying Microsoft for their software. As long as the Enterprise is licensed and using Microsoft, BitLocker is the recommended encryption service.

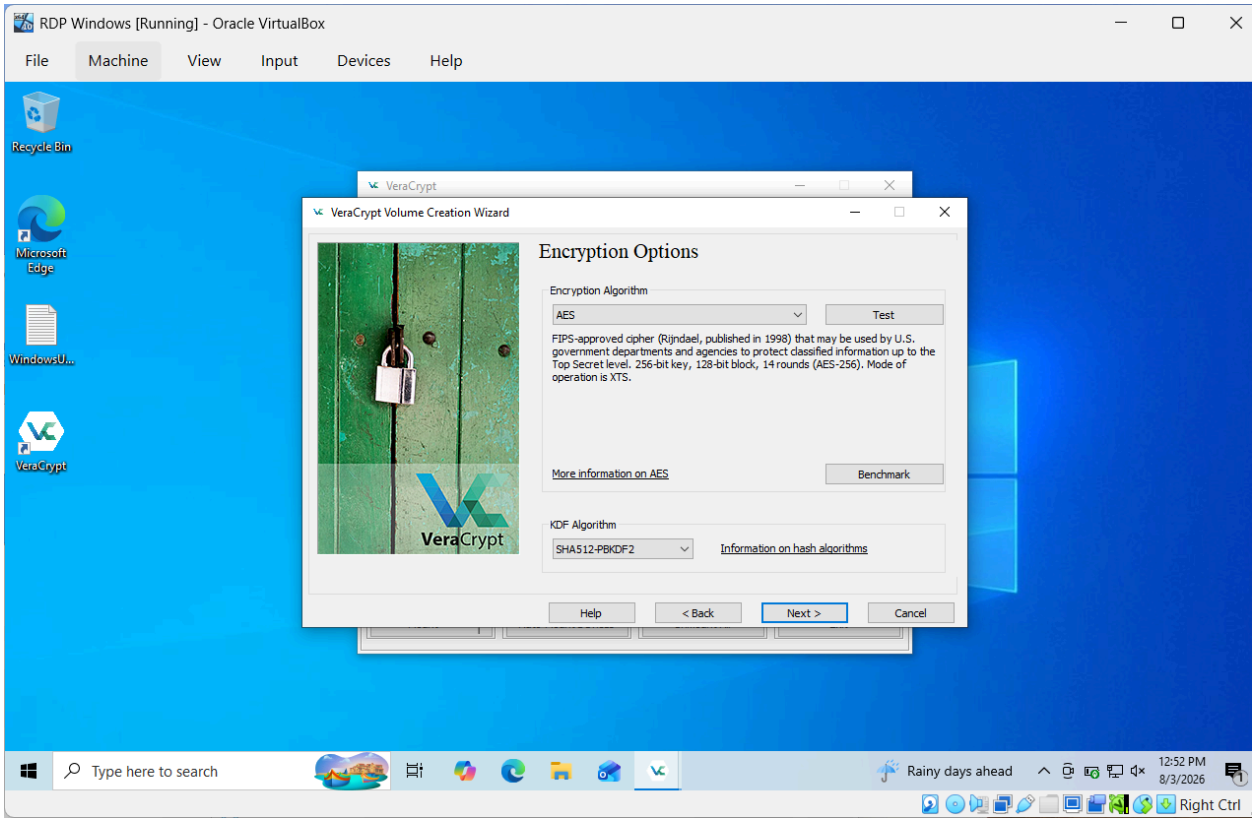


Creating the volume in VeraCrypt

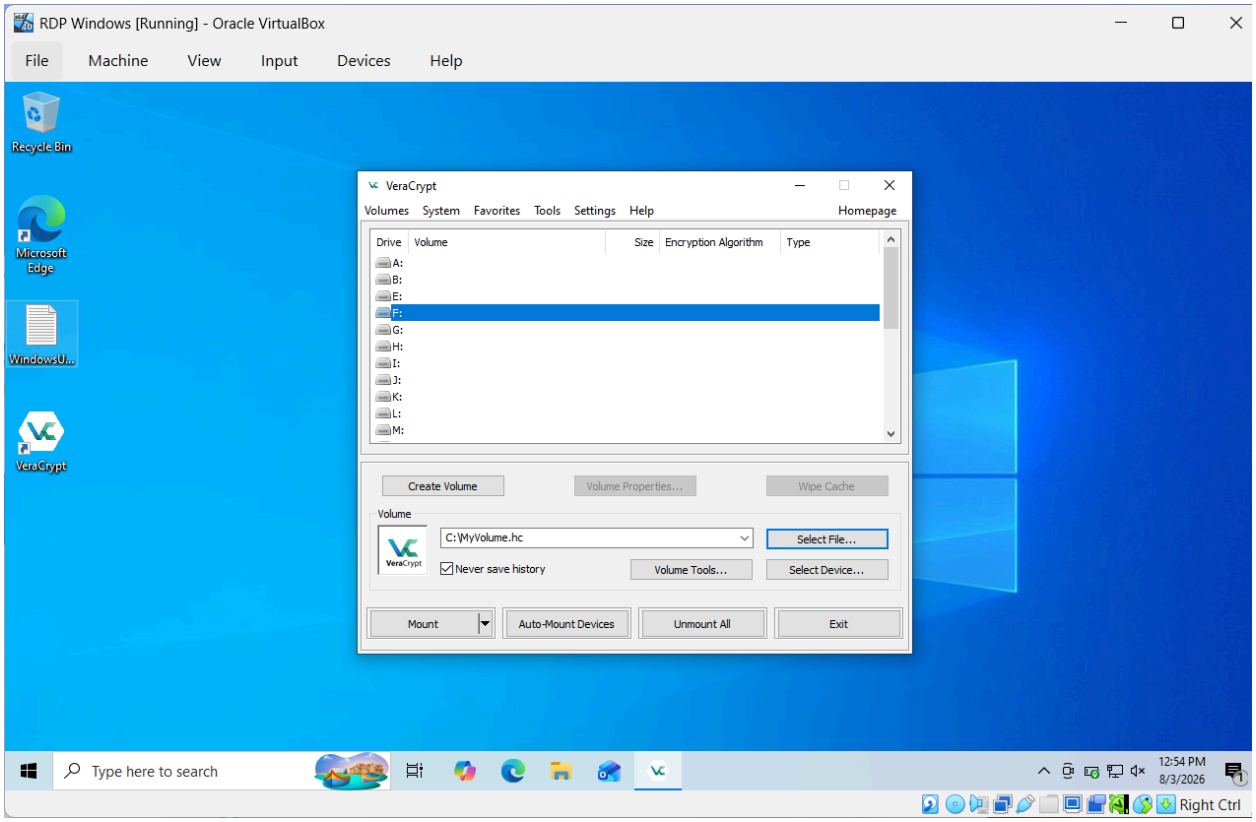

RDP Windows [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Recycle Bin
Microsoft Edge
WindowsU...
VeraCrypt
VeraCrypt
VeraCrypt Volume Creation Wizard
VeraCrypt Volume Creation Wizard
Create an encrypted file container
Creates a virtual encrypted disk within a file. Recommended for inexperienced users.
More information
Encrypt a non-system partition/drive
Encrypts a non-system partition on any internal or external drive (e.g. a flash drive). Optionally, creates a hidden volume.
Encrypt the system partition or entire system drive
Encrypts the partition/drive where Windows is installed. Anyone who wants to gain access and use the system, read and write files, etc., will need to enter the correct password each time before Windows boots. Optionally, creates a hidden system.
More information about system encryption
Help
< Back
Next >
Cancel
Type here to search
75°F Rain
12:51 PM
8/3/2026
Right Ctrl



Choosing the Disk location for the new volume.



Choosing the encryption method for the file. In this case, it's AES- SHA512.



Mounting the volume to “F:” drive and finishing up the folder encryption.